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Comparing Aquatic Environmental DNA Collection Methods Across a Diel Cycle in a Lotic System

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ABSTRACT

Environmental DNA (eDNA) sampling often requires a single sample collected at one point in time—a snapshot approach that provides limited information on biodiversity across temporal scales. To address this limitation, water samples were collected over 25 h from a single site on the Tangipahoa River in Louisiana, utilizing active and passive sampling methods. Active sampling involved collecting water samples from a predefined location at hourly intervals for 25 h, whereas passive sampling involved deploying a stationary filter for the entire duration. All water samples were filtered through glass microfiber filters, extracted, PCR-amplified targeting the 12S mtDNA region for fishes, and sequenced using Illumina technology following a pre-established protocol. The methods used for eDNA detection produced varying species richness metrics: active sampling identified 58 species, whereas passive sampling identified 37 species. On average, active sampling detected fewer species (mean 19, range 2–45) from a single hourly sample than the cumulative findings from passive sampling (totaling 37). Using a similarity clustering analysis, patterns associated with the time of day—civil twilight (daylight) versus dark (night)—showed significant differences ($p = 0.016$), although the distinction between the clusters was weak ($R\text{-stat} = 0.17$). These findings demonstrate that detection outcomes depend on sampling time and indicate that passive eDNA sampling integrates detections across time rather than representing a single sampling moment. This study illustrates the complexities of eDNA-based fish assemblage assessments and emphasizes the need for further research to optimize methodologies for accurate biodiversity monitoring.

1 | Introduction

A well-structured sampling design is a critical component of any research study, ensuring accuracy, reliability, validity, and comprehensiveness in the collected data (Bonar et al. 2024). In biodiversity assessments, the study design must consider numerous factors, such as the appropriate sampling time (e.g., day vs. night, or spring vs. summer), the sampling location, and the method of sampling (e.g., gear type) (Zale et al. 2013). Variations in an organism's mobility, morphology, and life-history can significantly affect its detectability during sampling (Villegas-Ríos et al. 2014). In fisheries management, traditional sampling methods are typically classified into two types: active

and passive (Zale et al. 2013). Active methods involve moving nets or gears (e.g., electrofishing, seines, trawls, dredges, etc.) to capture fish targets, while passive methods employ stationary nets or gears (e.g., trammel nets, hoop nets, trot lines) that do not require active movement (Zale et al. 2013). Whether active or passive, traditional methods are often selective for different fish sizes or species, and multiple gear types are necessary for a comprehensive fish assemblage assessment (Bonar et al. 2024).

The use of environmental DNA (eDNA) to study fish assemblages has increased due to advancements in technology for detecting and sequencing DNA, along with decreasing costs (Kelly et al. 2024; Lodge 2022). Specifically, eDNA is advantageous

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for managing invasive species, as eDNA can provide sensitive detection for some species relative to traditional sampling methods, which is crucial for preventing invasions (Ficetola et al. 2008; Jerde et al. 2019; Kelly et al. 2024; Oathout 2024). Furthermore, eDNA can evaluate multiple taxonomic levels or species through a technique called metabarcoding, which amplifies the sample DNA from a target region of interest and employs high-throughput sequencing (Kelly et al. 2024; Leray et al. 2013; Miya et al. 2015; Port et al. 2016; Yu et al. 2012).

Like traditional sampling methods, aquatic eDNA can be separated into active and passive collections, where active collection involves water filtration through a device, while passive collection relies on the hydrology of the system to move water through a filter. Given the time commitment required for active eDNA collection, passive eDNA collection methods offer a cost-effective, hands-off alternative. Passive techniques have been studied less extensively than active eDNA collection methods. Bessey et al. (2021) compared active and passive eDNA collection techniques in tropical and temperate marine systems and found passive eDNA collection comparable to active collection. Zhang et al. (2024) assessed active and passive filtering methods in subtropical marine waters in southeast China and discovered 85% and 59% similarities between the filtration types at two different localities. With further investigation across various aquatic systems, passive eDNA collection may reduce field effort and expand the application of eDNA in aquatic biomonitoring.

The collection of eDNA samples must be standardized, with protocol modifications and techniques changing constantly. However, while traditional monitoring methods (e.g., gillnets, electrofishing, mark-recapture) are well-developed and extensively studied (Murphy and Willis 1996; Nielsen and Johnson 1983; Zale et al. 2013), eDNA methods and techniques are still emerging and require further refinement across ecosystems and habitats. For instance, a single eDNA sample is typically collected from a location without considering the spatial or temporal variation in particular systems. Therefore, the objective of this study was to examine temporal variation in fish eDNA detections over a 25-h period and to evaluate whether passive sampling integrates detections occurring across time relative to hourly active sampling. We predicted that detections from active samples would vary across the diel cycle, whereas passive sampling would produce a temporally integrated detection signal relative to individual active samples.

2 | Materials and Methods

2.1 | Study Site

Sampling occurred in a freshwater lotic system, specifically the Tangipahoa River, in southeastern Louisiana, USA (Figure 1). The Tangipahoa River flows south-southeast for 177 km, beginning in southern Mississippi and crossing into southeastern Louisiana, where it continues to flow unobstructed into Lake Pontchartrain. The total river drainage basin covers about 2100 km² with an elevation change of around 100 m from the headwaters to the mouth (Mossa and Marks 2011). The majority of the land used in the basin is agricultural with

industries such as lumber and previously sand and gravel mining (Mossa and Marks 2011; Tangipahoa River, La, Entrance Channel 1970). In 2012, approximately 90 km of the river, extending from Louisiana's state line to the Interstate 12 crossing, was designated as part of the Natural and Scenic River System in Louisiana, established in 1970 (Boydston and Ethridge 2008; 2024 Louisiana Laws 2025). The designation prohibits channelization, channel realignment, clearing and snagging, impoundments, commercial clear-cutting within 30.48 m (100 ft) of the low water mark, and the use of motor vehicles to protect and preserve designated free-flowing rivers. The river drainage contains 72 fish species across 21 families, with the most common species being *Cyprinella venusta* (Blacktail shiner), *Lythrurus roseipinnis* (Cherryfin shiner), and *Gambusia affinis* (Mosquitofish) historically (Knight 1987). The mainstem of the Tangipahoa River contains 40 species, with the most common species being *Cyprinella venusta* (Blacktail shiner), *Lepomis macrochirus* (Bluegill), and *Lepomis solis* (Longear sunfish) contemporarily (LDWF 2022).

The site locally known as “Cherokee Beach” (30.602361, −90.437806) is located northeast of Hammond, Louisiana, USA, along LA Highway 442 (Figure 1). The maximum depth at this site does not exceed 1 m, and it is possible to wade across the entire 30 m width of the site. Large sandbars line the west bank, while hardwood trees and woody debris extend to the east bank. The geology and morphology of the Tangipahoa River are complex, featuring fluctuating seasonal water levels and diverse shoreline habitats. The Cherokee Beach site exemplifies this complexity, with upstream and downstream pools and runs deepening as the shoreline habitat transitions from swamp and cypress to sand and impervious structures. Meanders influence these transitions, and oxbow lakes can also overflow into the river during high water levels. This site was once a popular public swimming area and recreational launching point, but it is now privately owned.

2.2 | Active eDNA Collection

Before each sampling hour, all equipment was sterilized using a 10% bleach solution (7.5% NaOCl, The Clorox Company, Oakland, California, USA) allowed to air dry and then rinsed with 95% ethanol. Sampling took place over 25 consecutive hours, with one sample collected every hour from the center of the river. Sampling began at 1600 on 21 October 2022 and concluded at 1600 on 22 October 2022. The sampling location was marked with a PVC pole to ensure that water samples were taken from the exact spot during each hourly sampling event. Water samples were collected in the upper third of the water column using sterilized 500 mL Nalgene bottles every hour. Samples were obtained by wading into the river to waist height (~1 m). A sterilized 500 mL Nalgene bottle filled with deionized water was exposed to the air for 10 s at the sample site at the end of sampling (Krolow et al. 2022). The research was supported by an eDNA Collaborative microgrant, the Raney Fund Award from the American Society of Ichthyologists and Herpetologists, and internal sources. García-Machado et al. (2023). This served as the negative control for active and passive sampling to account for potential cross contamination. The negative control followed the same protocol as the active sample collections once it was

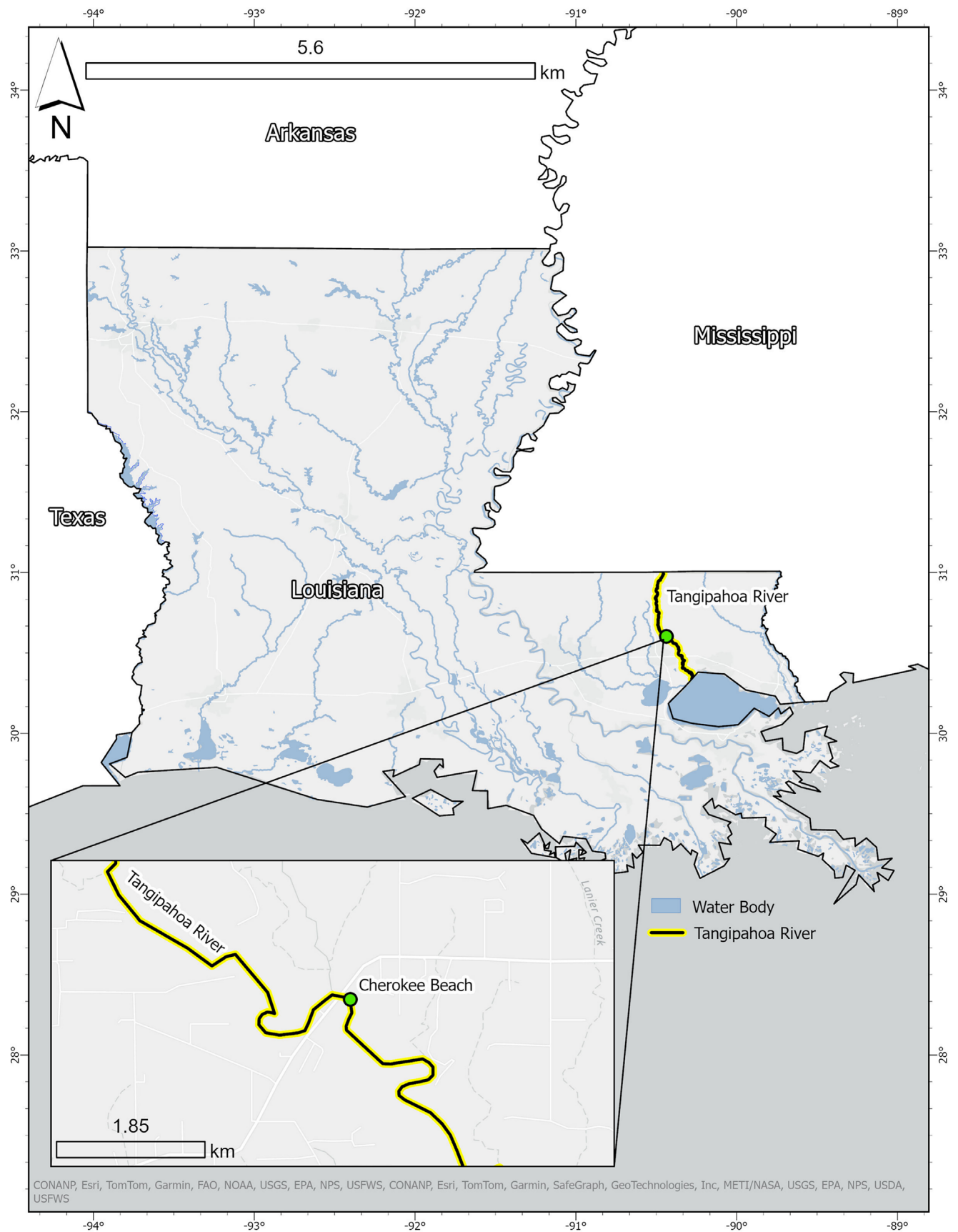


FIGURE 1 | Map of Louisiana, USA, showing the sampling site represented by the green point on the Tangipahoa River.

exposed to air. After collection, the 500 mL water samples were filtered through a 47 mm glass microfiber filter with a pore size of 1.5 μm (Whatman, catalog number 1827-047), using an open system vacuum pump (Welch by Gardner Denver, model number 2522B—01 A) and generator (Craftsman 3300i) on site immediately after collection. Each filter was placed in a 50 mL tube and stored on ice in a cooler (1–25 h) until it was transferred to a -20°C freezer in the lab, where it was stored until extraction. In addition to eDNA, abiotic parameters including time of collection, depth (meters, m), temperature ($^{\circ}\text{C}$), turbidity (nephelometric turbidity units, NTU), dissolved oxygen (milligrams/L, mg/L), percent dissolved oxygen (%DO), salinity (parts per thousand, ppt), and pH were also measured during each sampling event with a Hydro Lab Quanta water quality instrument (Hach Company, Loveland, Colorado, USA).

2.3 | Passive eDNA Collection

For the passive eDNA collection, three 47-mm glass microfiber filters with a pore size of 1.5 μm were mounted on a mesh screen in a PVC frame (Figure 2). The frame was positioned within three meters of the active eDNA collection site for the entire 25-h active sampling period and staked perpendicular to the flow and in the upper third of the water column. After 25 h, the filters were removed, placed in a 50 mL tube, and stored on ice in a cooler until being transferred to a -20°C freezer in the lab for storage until extraction.



FIGURE 2 | Passive environmental DNA collection frame featuring three 47-mm microfiber filters with a pore size of 1.5 μm .

2.4 | Extraction and DNA Amplification

All DNA extractions were conducted on a sterilized workbench using 10% bleach and 95% ethanol in a separate room from where DNA amplification occurred. The DNA extractions followed the modifications to the manufacturer's protocol for the Qiagen Power Water Kit (Qiagen Inc., catalog number 14900-100-NF) outlined in Krolow et al. (2022), and modified the incubation period slightly in the final step of the Qiagen protocol (described in Oathout [2024]). For samples with low DNA concentrations, below 1 ng/ μL , the following modifications were implemented: half the filter was sliced with a sterilized razor blade, the final ethanol wash step was extended to three and a half minutes, the EB Buffer (elution buffer) was allowed to stand for 1 min before completing the elution buffer step, and 50 μL of the elution buffer was used. Only one passive filter was extracted and sequenced due to funding. Sample DNA concentration was measured using the Qubit Flex Fluorometer (Thermo Fisher Scientific, catalog number Q33327). All samples were stored at -20°C until DNA amplification.

The 12S region of mitochondrial DNA was amplified using teleostean primers (MiFish-U-F and MiFish-U-R) developed by Miya et al. (2015). Amplification of this region requires a two-step polymerase chain reaction (PCR) approach. The first round of PCR amplifies the 12S region of interest from the extracted eDNA sample. The MiFish-U-F and MiFish-U-R primers also include overhanging adapter sequences for use in the second round of PCR. The reaction volume for the first round was 25 μL , comprising 1.25 μL of each primer at 10 μM , 6 μL to 8.5 μL of DNA template depending on DNA concentration, 2.5 μL to 4 μL of deionized water, and 12.5 μL of KAPA HiFi Hot Start ReadyMix (Kapa Biosystems Inc., Wilmington, Massachusetts, USA). Negative controls for PCR used 6 μL of deionized water in place of DNA template; the rest of the reaction stayed consistent. Each sample was amplified in triplicate under the following reaction conditions: initial denaturation at 98°C for 30 s, followed by 35 cycles of 98°C for 10 s, 61°C for 30 s, and 72°C for 20 s, concluding with a final extension at 72°C for 2 min. First-round PCR products were then visualized on a 1.5% agarose gel via gel electrophoresis run at 46 V alongside a 100 base pair ladder to verify the presence of targeted DNA. All field and PCR negative controls did not show the presence of targeted DNA and did not continue to the second round of PCR.

The second round of PCR adds barcodes for sample identification and adapter sequences into the targeted DNA region for Illumina sequencing. The template for this round consisted of a mix from the first-round triplicate. The reaction volume for the second round was 25 μL , which included 1.25 μL of each primer at 10 μM , 4 μL of DNA template, 6 μL of deionized water, and 12.5 μL of KAPA HiFi Hot Start ReadyMix. Each sample was amplified under the following reaction conditions: initial denaturation at 95°C for 3 min, followed by 12 cycles of 98°C for 20 s, 72°C for 30 s, concluding with a final extension at 72°C for 5 min. If samples initially failed to amplify in this second round of PCR, the following was utilized for a 25 μL reaction: 1.25 μL of each primer at 10 μM , 6 μL to 8 μL of DNA template, 4 μL to 2 μL of deionized water, and 12.5 μL of KAPA HiFi Hot Start ReadyMix. These samples adhered to the same reaction conditions. Prior to sequencing, samples were normalized to an even concentration and combined.

2.5 | Bioinformatics

The prepared eDNA library was sent to Azenta Genewiz (South Plainfield, New Jersey, USA) for sequencing. De-multiplexing, using the 8bp barcodes, was completed by Azenta Genewiz. The recovered Illumina sequencing data was analyzed using the MiFish pipeline on the MitoFish server (Sato et al. 2018). This pipeline initially performs quality checks on the sequences with FastQC and then utilizes DynamicTrim.pl. to remove 3' tails from sequences that have low-quality reads (Phred score < 10) (Sato et al. 2018). The default pipeline stated by Sato et al. (2018) merges pair-end reads using FLASH and removes erroneous reads containing uncalled nucleotides or with lengths of 229 ± 25 base pairs (Magoč and Salzberg 2011). TagCleaner removes primer sequences by allowing a maximum of 3 base pair mismatches (Schmieder et al. 2010). NCBI+ and Uclust are utilized for taxonomic assignment at the species level (Camacho et al. 2009; Edgar 2010). After the pipeline process, a dataset was generated featuring an 80%–96% similarity matrix and a $\geq 97\%$ similarity matrix. Analyses utilized the filtered threshold of the $\geq 97\%$ similarity matrix (Krolow et al. 2022). A minimum threshold of 0.1% of read counts per species per sample was applied, and species below this threshold were excluded (García-Machado et al. 2023). Additionally, marine fish species and non-fish species were removed. These filters were implemented to mitigate the possibility of false positives resulting from cross-contamination or tag jumping (Di Muri et al. 2020; Hänfling et al. 2016). Further filtering was performed to eliminate fish from the Tangipahoa River that had not been previously collected (Knight 1987; Piller, personal communication). The MitoFish Pipeline returned non-native species or species not previously reported in Louisiana, USA. In these cases, the second name assigned by the MitoFish Pipeline was used for analysis. Additionally, the MitoFish Pipeline identified three different *Micropterus* species. The analysis followed the nomenclature for these *Micropterus* species as described by Kim et al. (2022). Furthermore, the MitoFish Pipeline returned outdated nomenclature for *Lepomis megalotis* and *Aphredoderus sayanus*. This study follows the taxonomic revisions of these species as found in Kim et al. (2022) and Muller and Simons (2024).

2.6 | Data Analysis

All calculations were performed using R Statistical Software in Posit's RStudio Integrated Development Environment (v4.3.3; R Core Team 2024). The R packages vegan and phyloseq were utilized to calculate species richness, similarities, and conduct non-metric multidimensional scaling (NMDS) analysis (v3.20; McCune and Holmes 2013, v2.6.4; Oksanen et al. 2020, v6.1.1). Data visualization was achieved out using ggplot2 (v4.3.3; R Core Team 2024, v3.5.0; Wickham 2011). To address the study's first objective, species richness was calculated for each time interval, and similarity over time was determined using the Jaccard Index Matrix, as this method is suited for binary data (e.g., detected/not detected, presence/absence) and does not require abundances for calculating similarity (Jaccard 1912). An NMDS plot was generated to visualize the Jaccard Index Matrix, comparing sample period clusters. Sample period clusters represented the time from legal sunrise to legal sunset (0609 to 1723 on October 22 and 0610 to 1722 on October 23; hereafter, referred to as civil twilight) and from legal sunset to

legal sunrise (1723 to 0610 on October 22 to 23; hereafter, referred to as dark)—as determined by the US Navy Astronomical Applications Department. An analysis of similarity (ANOSIM) was performed to test for significance between the sampling periods (i.e., civil twilight versus dark). To further identify which species contribute to sample period clusters, the function `indval` in the package `labdsv` was employed (v.2.1-0, Roberts 2023). To address the study's second objective, species richness was calculated for each sampling method, and similarity across methods was assessed using the Jaccard Index Matrix. The Jaccard Index Matrix was also used to compare the species richness of this study's eDNA methods and the traditional sampling from Knight (1987), the most recent fish assemblage investigation for the Tangipahoa River. Finally, a linear model was used to examine the effects of time on species richness and on the potential degradation of DNA during field sample storage.

3 | Results

3.1 | Abiotic Data

During the sampling period, depth and salinity remained stable, consistently measuring 0.91 m and 0.03 ppt, respectively. Similarly, temperature and pH showed no variation, with average values of 18.26°C (± 0.29 ; 95% CI) and 6.65 (± 0.17), respectively. As anticipated, flow, turbidity, and dissolved oxygen exhibited minor changes throughout the sampling period (Figure S2). Flow, turbidity, and dissolved oxygen ranged from 1.67 to 3.44 m/s, 21.2 to 91.0 NTU, and 1.52 to 3.38 mg/L, respectively. Concentration of DNA across sampling events did not change significantly ($F\text{-stat}_{1,24} = 2.88$, $p\text{ value} = 0.10$).

3.2 | Sequence Data

A total of 2,699,670 filtered reads were obtained from the 26 samples (25 active and 1 passive) collected at the Tangipahoa River Cherokee Beach site. The average number of reads per sample was 103,834. The tenth sampling hour (0200) yielded 1,409,803 reads, the highest among all the samples, while the eighth sampling hour (0000) recorded the fewest reads, totaling 43. The species with the highest read count across all samples was *Moxostoma poecilurum* (Blacktail Redhorse), with 396,976 reads. In contrast, *Hybognathus nuchalis* (Mississippi Silvery Minnow) displayed the lowest read count among all samples with only 16 reads. The average number of reads per species was 45,757. Rarefaction curves of read abundances for the detected fishes did not indicate adequate sequencing depths for all samples (Figure S1).

3.3 | BLAST Misidentifications

Six non-native species in Louisiana were identified through passive and active eDNA sampling. These species were *Ictalurus meridionalis* (Southern Blue Batfish), *Lepisosteus platyrhincus* (Florida Gar), *Micropterus floridanus* (Florida Bass), *Micropterus salmoides salmoides*, *Micropterus treculii* (Guadalupe Bass), *Etheostoma zonale* (Banded Darter), and *Pimephales tenellus* (Slim Minnow). The second names assigned to these species in the Mitofish Pipeline were

Micropterus nigricans (Largemouth Bass), *Ictalurus furcatus* (Blue Catfish), *Lepisosteus oculatus* (Spotted Gar), *Micropterus punctulatus* (Spotted Bass), *Etheostoma lynceum* (Bryteye Darter), and *Pimephales vigilax* (Bullhead Minnow), respectively, all species found in the Tangipahoa River (Knight 1987). Two species were identified from Louisiana but are not known to occur in the Tangipahoa River basin. *Erimyzon oblongus* was identified; however, the representative sequence for this species was returned as a 100% match to *Erimyzon tenuis* when using the NCBI Blast database. *Cyprinella lutrensis* was identified; however, *Notropis volucellus* was the next match when using the NCBI Database and is found in the Tangipahoa River basin. Furthermore, with recent taxonomic revisions, the assignment *Aphredoderus sayanus* became *Aphredoderus gibbosus* and *Lepomis megalotis* became *Lepomis solis* (Kim et al. 2022; Muller and Simons 2024).

3.4 | Species Richness

A total of 1534 individuals (molecular operational taxonomic units, MOTUs) were detected throughout the study, representing 59 different species (Table 1; Figure 3). Active sampling detected 58 species, with 22 species unique to this method (Table 1). The lowest number of detections for active sampling was two species during sample hours 8 (0000) and 9 (0100). The highest number of detections for active sampling was 45 species during sample hour 24 (1500). The species richness across active sampling events did not change significantly ($F\text{-stat}_{1,24} = 0.02$, p value = 0.86). Passive sampling detected 37 species, with *Noturus leptacanthus* (Speckled Madtom) being the only species unique to this method (Table 1). Nineteen different families of fish were detected at the Cherokee Beach site in the Tangipahoa River. The most abundant family was Centrarchidae, detected in 19.7% of all samples, followed by Catostomidae (13.1%), Leuciscidae (13.1%), and Ictaluridae (9.8%). The most abundant genus was *Lepomis*, detected in 13.1% of all samples, followed by *Percina* (6.6%), *Notropis* (4.9%), and *Etheostoma* (4.9%).

Passive and active sampling methods yielded distinct species richness metrics. Active sampling identified 58 species, averaging approximately 19 species per sampling event, whereas passive sampling detected 37 species. The Jaccard Similarity index between these sampling methods was 0.61, indicating a 61% similarity in the species detected. The species accumulation curve is near asymptotic and illustrates adequate sampling of fishes using the active eDNA sampling method (Figure 4).

3.5 | Fish Assemblages

Active sampling detected 58 species, whereas passive sampling detected 37 species, with an overlap of 0.61 (Jaccard similarity). Non-metric multidimensional scaling did not show clear clustering by sampling method. However, detections differed weakly but significantly between civil twilight and dark sampling periods (ANOSIM p value = 0.016, $R = 0.17$; Figure 5). Four species were detected only during civil twilight, and five only during dark sampling periods (Table 2). All species and their relative frequencies for each sampling period are provided in Table S1.

The Jaccard similarity between this study and the five sampling days conducted by Knight (1987) in the Cherokee Beach area was 0.19 and not statistically significant. Similarly, the similarity between combined eDNA detections (passive and active) and Knight's (1987) basin-wide dataset was 0.45, which was not statistically significant.

4 | Discussion

This study examined temporal variation in fish assemblage detections over a 25-h diel cycle using active (hourly water filtration) and passive (time-integrated filter) eDNA sampling at a single lotic site. Active sampling detected 58 species with high hourly variability (2–45 species per sample), whereas passive sampling detected 37 species and produced a temporally integrated assemblage signal. Assemblage composition differed significantly but weakly between daylight (civil twilight) and nighttime (dark) sampling periods. These results indicate that eDNA collected as a single snapshot does not necessarily represent the assemblage present across a diel cycle, and that passive sampling integrates detections occurring over time rather than reflecting a single moment in occupancy.

Accurately estimating species richness and assemblage structure remains challenging because detection probability varies across species and sampling methods (Gamito et al. 2012; Pompeu et al. 2021). Determining how much sampling effort is required to adequately characterize assemblages is, therefore, a central question in ecological monitoring (Kéry and Schmidt 2008). Our results demonstrate that temporal sampling design is a major component of that effort when using eDNA in flowing systems. Rather than reflecting a fixed assemblage at a location, hourly eDNA detections varied substantially across the diel cycle, indicating that detection depends on the sampling time.

Environmental DNA approaches are still developing compared to decades of research on traditional sampling methods. Previous work has focused on optimizing sample volume (Cantera et al. 2019), filter type and pore size (Bessey et al. 2022), and primer set selection (Thomsen et al. 2012). The present study complements that literature by providing unique insights into how the temporal sampling strategy itself influences fish assemblage characterization.

4.1 | Diel Patterns

The diel differences observed in this study likely reflect fish behavior and the transport of eDNA within a lotic system. Environmental DNA collected at a point in a river originates not only from locally present individuals but also from organisms upstream. Previous studies have documented downstream transport of eDNA over kilometers in river systems and how biological processes—such as feeding and activity patterns—influence DNA shedding rates (Wilcox et al. 2016; García-Machado et al. 2023). For example, Laporte et al. (2021) demonstrated transport of DNA up to 6 km before detecting an assemblage shift using eDNA metabarcoding. Additionally, (Deiner and Altermatt 2014) found eDNA transport of two invertebrate species in a river system up

TABLE 1 | All fish species detected at the Cherokee Beach site (30.602361, -90.437806) in the Tangipahoa River, Louisiana, USA. A '1' denotes species detected and a '0' denotes species not detected using active or passive aquatic eDNA sampling. Water samples were collected over 25 h from a single point. Active sampling involved collecting water samples from a predefined location at hourly intervals, whereas passive sampling involved deploying a stationary filter for the entire duration.

Family	Species	Active sample	Passive sample
Lepisosteidae	<i>Lepisosteus oculatus</i>	1	0
	<i>Lepisosteus platostomus</i>	1	1
Anguillidae	<i>Anguilla rostrata</i>	1	1
Clupeidae	<i>Dorosoma cepedianum</i>	1	1
	<i>Dorosoma petenense</i>	1	1
Cyprinidae	<i>Cyprinus carpio</i>	1	0
Leuciscidae	<i>Cyprinella venusta</i>	1	1
	<i>Hybognathus nuchalis</i>	1	0
	<i>Hybopsis amblops</i>	1	0
	<i>Luxilus chrysocephalus</i>	1	1
	<i>Notropis longirostris</i>	1	1
	<i>Notropis texanus</i>	1	1
	<i>Notropis volucellus</i>	1	1
	<i>Pimephales vigilax</i>	1	0
	<i>Carpionodes carpio</i>	1	1
	<i>Cycleptus elongatus</i>	1	0
	<i>Erimyzon tenuis</i>	1	1
	<i>Hypentelium nigricans</i>	1	1
	<i>Ictiobus cyprinellus</i>	1	1
Catostomidae	<i>Ictiobus niger</i>	1	0
	<i>Minytrema melanops</i>	1	0
	<i>Moxostoma poecilurum</i>	1	1
	<i>Ameiurus natalis</i>	1	1
	<i>Ictalurus furcatus</i>	1	1
	<i>Ictalurus punctatus</i>	1	1
	<i>Noturus gyrinus</i>	1	1
	<i>Noturus leptacanthus</i>	0	1
	<i>Pylodictis olivaris</i>	1	1
	<i>Esox americanus vermiculatus</i>	1	1
	<i>Aphredoderus gibbosus</i>	1	0
	<i>Mugil cephalus</i>	1	1
	<i>Labidesthes sicculus</i>	1	0
Atherinopsidae	<i>Membras martinica</i>	1	0
	<i>Menidia audens</i>	1	1
	<i>Fundulus chrysotus</i>	1	0
Fundulidae	<i>Fundulus olivaceus</i>	1	1

(Continues)

TABLE 1 | (Continued)

Family	Species	Active sample	Passive sample
Poeciliidae	<i>Gambusia affinis</i>	1	1
Centrarchidae	<i>Ambloplites ariommus</i>	1	1
Centrarchidae cont.	<i>Lepomis cyanellus</i>	1	0
	<i>Lepomis gulosus</i>	1	0
	<i>Lepomis humilis</i>	1	0
	<i>Lepomis macrochirus</i>	1	1
	<i>Lepomis marginatus</i>	1	1
	<i>Lepomis microlophus</i>	1	0
	<i>Lepomis miniatus</i>	1	0
	<i>Micropterus nigricans</i>	1	1
	<i>Micropterus punctulatus</i>	1	1
	<i>Lepomis solis</i>	1	1
	<i>Pomoxis annularis</i>	1	1
Percidae	<i>Ammocrypta beanii</i>	1	1
	<i>Etheostoma histrio</i>	1	1
	<i>Etheostoma lynceum</i>	1	1
	<i>Etheostoma swaini</i>	1	0
	<i>Percina nigrofasciata</i>	1	1
	<i>Percina sciera</i>	1	0
	<i>Percina suttkusi</i>	1	0
	<i>Percina vigil</i>	1	0
Sciaenidae	<i>Aplodinotus grunniens</i>	1	1
Achiridae	<i>Trinectes maculatus</i>	1	0

to 10km downstream of their source populations, after which it declined below their detection threshold. When investigating the effects of feeding on eDNA shedding rates in carp, Klymus et al. (2015) found a significant increase in the amount of eDNA shed. Consequently, hourly active samples likely capture transient detection conditions rather than a static assemblage.

The clustering patterns by time of day—civil twilight denotes daylight hours, and dark denotes nighttime hours—show weak qualitative distinctions but remain significantly different. Therefore, a single daytime snapshot sample may lead to an incomplete interpretation of assemblage detections across a diel period. In coastal regions of Denmark, Jensen et al. (2022) detected diel shifts in fish assemblages using active eDNA sampling approaches. Additionally, traditional sampling methods have demonstrated diel shifts in fish assemblages in lotic systems. For example, Arrington and Winemiller (2003) found differences between diurnal and nocturnal fish assemblages when using seines and cast nets in a neotropical river, and Pierce et al. (2001) detected significantly greater species richness at night than during the day for multiple gear types in two lentic systems. Our results support

the idea that employing eDNA in a snapshot manner could lead to incomplete interpretation of assemblage detections.

4.2 | Traditional Sampling Methods and eDNA

Comparisons with conventional surveys must be interpreted in the context of sampling design. Knight (1987) reported 72 fish species across 21 families from 59 sites and 85 sampling events in the Tangipahoa River basin (specifically the river and lotic tributaries), including five sampling days in the Cherokee Beach area. Knight (1987) used multiple capture gears (backpack electrofishers, seines, gillnets) over 18 months, from 1985 to 1986. However, our study examined a single location continuously over a 25-h period using molecular detection. The low similarity between studies likely reflects differences in detection processes rather than disagreement in assemblage composition. These comparisons, therefore, reflect differences in observation methods and detection probabilities rather than differences in the true fish community present. Capture gears can detect only individuals physically present during sampling, whereas

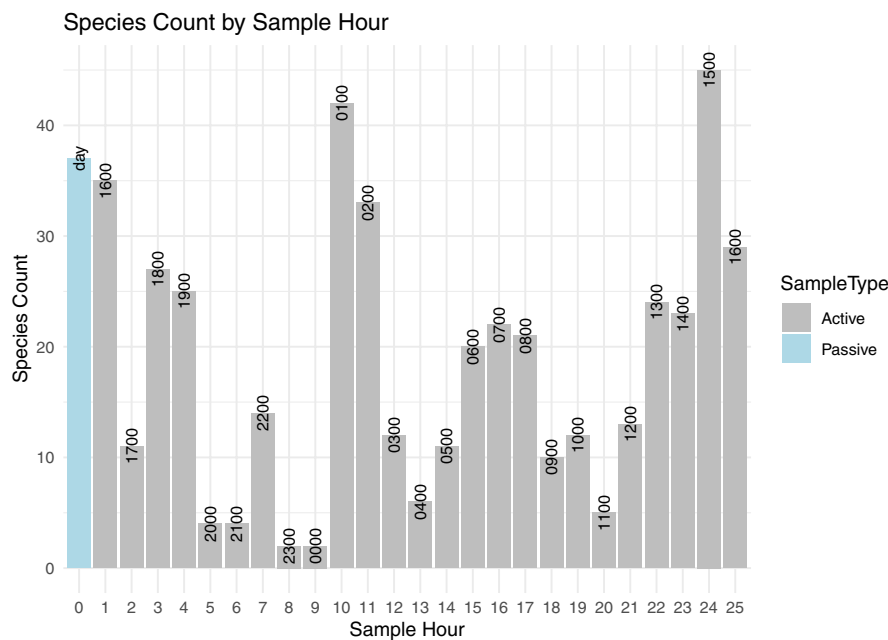


FIGURE 3 | The count of species detected per hour of active sampling (gray bars) and the number of species detected through passive sampling (light blue bar, sample hour 0) at the Cherokee Beach site. Sample hour 1 indicates the first sample taken at 1600 on 21 October 2022, while sample hour 25 represents the last sample collected at 1600 on 22 October 2022. Sample hour 0 represents the passive sample.

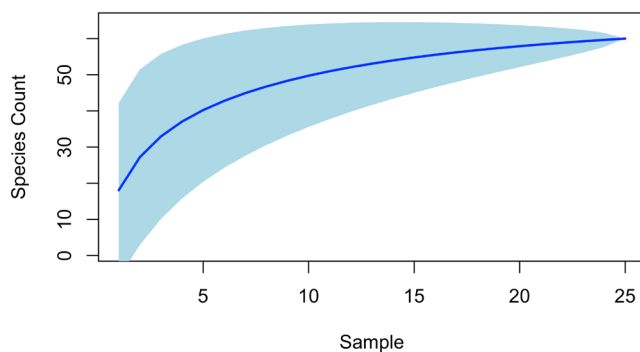


FIGURE 4 | Estimated species accumulation curve (dark blue line) and 95% confidence intervals (light blue fill) of active hourly samples, with the number of samples represented on the x-axis and the species count on the y-axis.

eDNA may detect species moving through the area or located upstream. Previous comparisons between eDNA and traditional sampling have found that eDNA often detects additional taxa in lotic systems, particularly smaller-bodied or rare species (Hallam et al. 2021; Piggott et al. 2021; Robinson et al. 2019).

4.3 | Active Versus Passive eDNA Sampling

Active and passive eDNA sampling measure different ecological signals. Passive sampling integrates eDNA detection signals over time, whereas hourly active samples represent instantaneous detection conditions. In this study, passive sampling detected fewer total species than the combined active sampling effort but more than a typical single active sample, indicating that passive collection accumulates signals produced across multiple behavioral states and time periods. These results are intriguing as they

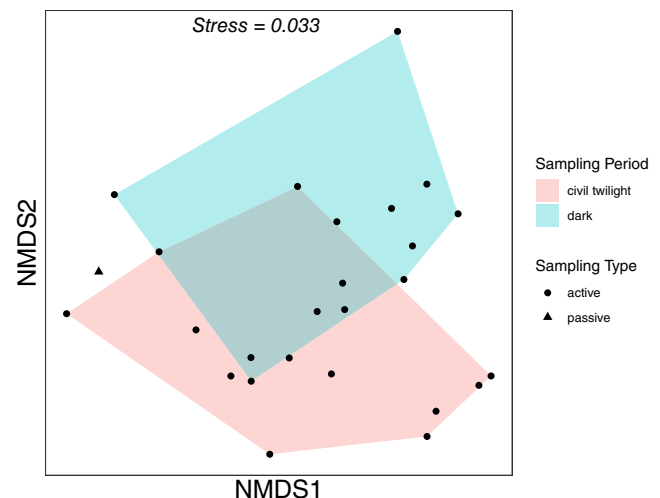


FIGURE 5 | Nonmetric Multidimensional Scaling plot illustrating the comparison of fish assemblages between sampling methods over time. Clusters are based on the sample collection period (civil twilight [pink], dark [blue]) and sampling type (active [circles], passive [triangle]). Significant similarities between sampling periods were not identified (Jaccard value = 0.59). Note that the passive sampling triangle is not nested within either active sampling period cluster.

challenge the notion that passive and active sampling techniques would produce similar assemblage clusters and species richness values—a hypothesis supported in the literature. For instance, Chen et al. (2024) observed similar fish assemblages and species richness between active filtration (500 mL/h; same as this study) and a 25-h passive eDNA sampler. In shallow (<20m) temperate marine systems, Bessey et al. (2021) found similar species richness across active and passive eDNA sampling methods but noted significantly distinct species composition rates depending on the

TABLE 2 | All fish species detected at the Cherokee Beach site (30.602361, −90.437806) in the Tangipahoa River, Louisiana, USA, using active and passive eDNA sampling methods. Water samples were collected over 25 h from a single point. Active sampling involved collecting water samples from a predefined location at hourly intervals, whereas passive sampling involved deploying a stationary filter for the entire duration. Individual species are separated by their association with passive or active sampling and civil twilight (0609 to 1723 on October 22 and 0610 to 1722 on October 23) or dark (1723 to 0610 on October 22 to 23). These times reflect legal sunrise to legal sunset and vice versa, respectively, as determined by the US Navy Astronomical Applications Department.

Sampling method	Sampling period	Species
Active	Civil Twilight	<i>Pimephales vigilax</i>
Active	Dark	<i>Ictiobis niger</i>
Active cont.	Dark cont.	<i>Percina sciera</i>
		<i>Lepisosteus oculatus</i>
		<i>Lepomis cyanellus</i>
		<i>Lepomis microlophus</i>
		<i>Lepomis miniatus</i>
		<i>Membra martinica</i>
		<i>Minytrema melanops</i>
Passive	Day	<i>Anguilla rostrata</i>
		<i>Dorosoma cepedianum</i>
		<i>Etheostoma lynceum</i>
		<i>Ictiobus cyprinellus</i>
		<i>Fundulus olivaceus</i>
		<i>Moxostoma poecilurum</i>
		<i>Cyprinella venusta</i>
		<i>Lepomis macrochirus</i>
		<i>Ictalurus punctatus</i>
		<i>Lepomis marginatus</i>
		<i>Pylodictis olivaris</i>
		<i>Mugil cephalus</i>
		<i>Aplodinotus grunniens</i>
		<i>Micropterus punctulatus</i>
		<i>Etheostoma histrio</i>
		<i>Micropterus nigricans</i>
		<i>Lepomis solis</i>
		<i>Notropis longirostris</i>
		<i>Notropis volucellus</i>
		<i>Menidia audens</i>
		<i>Gambusia affinis</i>
		<i>Ammocrypta beanie</i>
		<i>Hypentelium nigricans</i>
		<i>Carpoides carpio</i>
		<i>Ambloplites ariommus</i>
		<i>Lepisosteus platostomus</i>
		<i>Dorosoma petenese</i>
		<i>Pomoxis annularis</i>

passive collection filter and site. Oppositely, Stevens et al. (2024) found that active eDNA collection outperformed passive eDNA collection over a 4-h period. Although these three studies cannot be directly compared with the present study due to differing methodologies and habitats, the varied results show a need for further investigation into eDNA sampling approaches to better standardize eDNA sampling alongside traditional methods.

The identification of six non-native species not previously reported in the Tangipahoa River basin suggests that, although reference sequences are available in GenBank and MiFish, geographic variability in the 12S locus may result in species misidentification. These misidentifications indicate that data produced by eDNA sequencing still require vetting, as done in this study. Fortunately, the second scientific name for each species listed in the Mitofish Pipeline is native and has been previously documented in the Tangipahoa River. As a result, the findings from the active and passive eDNA sampling datasets were obtained without concern for species misidentification.

4.4 | Abiotic Impact

Abiotic conditions can influence eDNA persistence, transport, and detection. Temperature (Eichmiller et al. 2016; Lamb et al. 2022; McCartin et al. 2022), dissolved oxygen (Barnes et al. 2021), and pH (Lance et al. 2017) affect degradation rates, while turbidity and flow influence transport and dilution (Collins et al. 2018; Strickler et al. 2015; Zhao et al. 2023). Although diel changes in assemblage detections were observed, hourly variation in measured water-quality variables did not correspond to consistent changes in species richness in this dataset. Our results suggest that behavioral activity and transport processes likely contributed more to the observed temporal differences than short-term abiotic fluctuations.

A limitation of this study is that field and laboratory negative controls were evaluated by electrophoresis but were not sequenced through the full bioinformatic pipeline. Although no amplification was visually detected, low-level contamination can occur in metabarcoding workflows below visible thresholds. Therefore, we cannot completely exclude the possibility of trace contamination. However, detections were consistent with known regional fauna, and implausible marine or geographically distant taxa were removed during taxonomic curation, reducing the likelihood that contamination influenced interpretation. Future eDNA studies should include negative controls in sequencing and bioinformatics processing to better quantify background signal and contamination risk.

Overall, because diel differences were present, a single sampling time would not fully represent assemblage composition. If limited to a single sampling period, a crepuscular sampling window would likely yield a more representative snapshot than strictly daytime sampling, although repeated sampling remains preferable.

4.5 | Future Directions

This study emphasizes the complexities of eDNA-based fish assemblage assessments and highlights the need for further research to optimize methodologies for accurate biodiversity monitoring.

Although the findings indicate diel variability in species detection, further studies across diverse aquatic environments are essential to evaluate whether these patterns hold among ecosystems. Future research should explore how diel cycles interact with habitat complexity, seasonal variations, and hydrodynamic processes to influence eDNA dispersal and persistence. Long-term monitoring incorporating eDNA sampling across temporal scales—from daily to seasonal and annual—could offer deeper insights into species turnover and shifts in assemblage composition.

Moreover, the differences observed between active and passive eDNA sampling methods underscore the need for standardized methodologies. Here we detected 61% similarity between the two methods. Potential buildup of particles and sediment leading to inhibition on the passive filter could be an explanation. Another is the difference in water volume filtered between methods. This study collected 500 mL of water for the active sample while others have collected a liter or more (Cantera et al. 2019; Bessey et al. 2021; Hallam et al. 2021; Jensen et al. 2022). By comparison, in this study with the average stream flow and size of the filter, over an hour, thousands of liters are passing through the passive filter. Future research should focus on the optimal collection volume, deployment duration, filter types, and molecular techniques for active and passive eDNA sampling to improve consistency and facilitate cross-study comparisons. Understanding daily shedding rates for individual species is also essential for refining eDNA detection methods. Furthermore, broadening comparative analyses with traditional sampling methods will help clarify the role of eDNA as either a complementary or standalone tool in ecological monitoring. Since environmental factors such as temperature, pH, dissolved oxygen, and turbidity may affect eDNA detection and decay, future investigations should integrate abiotic variables into predictive models to improve the accuracy of fish assemblage and species richness estimates. Advancements in reference databases will also be crucial for minimizing species misidentification and enhancing taxonomic resolution. Ultimately, incorporating these insights into conservation and management frameworks will bolster eDNA's utility in monitoring aquatic ecosystems and assessing biodiversity.

5 | Conclusions

This study indicates that a single eDNA sample may not adequately represent temporal variation in detections within a fish assemblage. If a more representative characterization of detected assemblage composition using eDNA is the goal, multiple active eDNA samples should be collected throughout the day within a specific study area. As demonstrated by Jensen et al. (2022), eDNA can quickly recover multiple assemblages. The reasons for these distinct community assemblages can also be linked to various factors affecting eDNA beyond fish activity patterns, such as abiotic variables (e.g., temperature, salinity, turbidity), chlorophyll levels in the water, and habitat types (García-Machado et al. 2021; Laporte et al. 2021; Port et al. 2016). Furthermore, the findings from this study suggest that a passive eDNA sampler may provide a practical alternative to multiple active eDNA samples in similar lotic conditions. The required submersion time for the passive sampler may be shorter than the 25-h submersion period reported in the current study, as previous research indicates that extended submersion

time does not yield higher estimates of species richness (Bessey et al. 2021, 2022; Kirtane et al. 2020). There remain knowledge gaps in understanding eDNA dynamics across aquatic systems and their ability to accurately detect fish over time. Through further validation of eDNA and standardization of methodologies, eDNA can evolve into a more cost-effective and time-efficient tool for assessing fish diversity across large temporal and spatial scales (Eichmiller et al. 2016; García-Machado et al. 2021; Lamb et al. 2022; Takahara et al. 2012; Thomas et al. 2018).

Author Contributions

C.E.O.: conception of the study, acquisition, analysis and interpretation of data, writing of the manuscript. C.E.G.: acquisition, analysis and interpretation of data. T.S.C.: acquisition, analysis and interpretation of data, writing of the manuscript. K.R.P.: conception of the study, acquisition, analysis and interpretation of data, writing of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Raw sequence data supporting this study's findings are available at the NCBI Sequence Read Archive (SRA), reference number PRJNA1244917, after the acceptance of this manuscript. Other data and code supporting this study's findings are available upon request.

References

- 2024 Louisiana Laws: Revised Statutes: Title 56—Wildlife and Fisheries: §56:1847. 2025. *Natural and Scenic Rivers*. Justia Law. <https://law.justia.com/codes/louisiana/revised-statutes/title-56/rs-56-1847/>.
- Arrington, D. A., and K. O. Winemiller. 2003. "Diel Changeover in Sandbank Fish Assemblages in a Neotropical Floodplain River." *Journal of Fish Biology* 63, no. 2: 442–459. <https://doi.org/10.1046/j.1095-8649.2003.00167.x>.
- Barnes, M. A., W. L. Chadderton, C. L. Jerde, A. R. Mahon, C. R. Turner, and D. M. Lodge. 2021. "Environmental Conditions Influence eDNA Particle Size Distribution in Aquatic Systems." *Environmental DNA* 3, no. 3: 643–653. <https://doi.org/10.1002/edn3.160>.
- Bessey, C., Y. Gao, Y. B. Truong, H. Miller, S. N. Jarman, and O. Berry. 2022. "Comparison of Materials for Rapid Passive Collection of Environmental DNA." *Molecular Ecology Resources* 22, no. 7: 2559–2572. <https://doi.org/10.1111/1755-0998.13640>.

- Bessey, C., S. Neil Jarman, T. Simpson, et al. 2021. "Passive eDNA Collection Enhances Aquatic Biodiversity Analysis." *Communications Biology* 4, no. 1: 236. <https://doi.org/10.1038/s42003-021-01760-8>.
- Bonar, S. A., J. D. Conroy, S. Contreras-Balderas, and A. C. Iles. 2024. "Chapter 1: An Introduction to Standardized Sampling." In *Standard Methods for Sampling North American Freshwater Fishes*, 2nd ed., 1–22. American Fisheries Society.
- Boydston, J., and B. Ethridge. 2008. "US Environmental Protection Agency."
- Camacho, C., G. Coulouris, V. Avagyan, et al. 2009. "BLAST+: Architecture and Applications." *BMC Bioinformatics* 10, no. 1: 421. <https://doi.org/10.1186/1471-2105-10-421>.
- Cantera, I., K. Cilleros, A. Valentini, et al. 2019. "Optimizing Environmental DNA Sampling Effort for Fish Inventories in Tropical Streams and Rivers." *Scientific Reports* 9, no. 1: 3085. <https://doi.org/10.1038/s41598-019-39399-5>.
- Chen, X., S. Li, J. Zhao, and M. Yao. 2024. "Passive eDNA Sampling Facilitates Biodiversity Monitoring and Rare Species Detection." *Environment International* 187: 108706. <https://doi.org/10.1016/j.envint.2024.108706>.
- Collins, R. A., O. S. Wangensteen, E. J. O'Gorman, S. Mariani, D. W. Sims, and M. J. Genner. 2018. "Persistence of Environmental DNA in Marine Systems." *Communications Biology* 1, no. 1: 1–11. <https://doi.org/10.1038/s42003-018-0192-6>.
- Deiner, K., and F. Altermatt. 2014. "Transport Distance of Invertebrate Environmental DNA in a Natural River." *PLoS One* 9, no. 2: e88786. <https://doi.org/10.1371/journal.pone.0088786>.
- Di Muri, C., L. L. Handley, C. W. Bean, et al. 2020. "Read Counts From Environmental DNA (eDNA) Metabarcoding Reflect Fish Abundance and Biomass in Drained Ponds." *bioRxiv*. <https://doi.org/10.1101/2020.07.29.226845>.
- Edgar, R. C. 2010. "Search and Clustering Orders of Magnitude Faster Than BLAST." *Bioinformatics* 26, no. 19: 2460–2461. <https://doi.org/10.1093/bioinformatics/btq461>.
- Eichmiller, J. J., S. E. Best, and P. W. Sorensen. 2016. "Effects of Temperature and Trophic State on Degradation of Environmental DNA in Lake Water." *Environmental Science & Technology* 50, no. 4: 1859–1867. <https://doi.org/10.1021/acs.est.5b05672>.
- Ficetola, G. F., C. Miaud, F. Pompanon, and P. Taberlet. 2008. "Species Detection Using Environmental DNA From Water Samples." *Biology Letters* 4, no. 4: 423–425. <https://doi.org/10.1098/rsbl.2008.0118>.
- Gamito, R., S. Pasquaudo, A. Courrat, et al. 2012. "Influence of Sampling Effort on Metrics of Fish-Based Indices for the Assessment of Estuarine Ecological Quality." *Ecological Indicators* 23: 9–18. <https://doi.org/10.1016/j.ecolind.2012.03.009>.
- García-Machado, E., M. Laporte, E. Normandeau, et al. 2021. "Fish Community Shifts Along a Strong Fluvial Environmental Gradient Revealed by eDNA Metabarcoding." *Environmental DNA* 4: 117–134. <https://doi.org/10.1002/edn3.221>.
- García-Machado, E., E. Normandeau, G. Côté, and L. Bernatchez. 2023. "How eDNA Data Filtration, Sequence Coverage, and Primer Selection Influence Assessment of Fish Communities in Northern Temperate Lakes." *Environmental DNA* 5, no. 6: 1216–1233. <https://doi.org/10.1002/edn3.444>.
- Hallam, J., E. L. Clare, J. I. Jones, and J. J. Day. 2021. "Biodiversity Assessment Across a Dynamic Riverine System: A Comparison of eDNA Metabarcoding Versus Traditional Fish Surveying Methods." *Environmental DNA* 3, no. 6: 1247–1266. <https://doi.org/10.1002/edn3.241>.
- Hänfling, B., L. Lawson Handley, D. S. Read, et al. 2016. "Environmental DNA Metabarcoding of Lake Fish Communities

- Reflects Long-Term Data From Established Survey Methods.” *Molecular Ecology* 25, no. 13: 3101–3119. <https://doi.org/10.1111/mec.13660>.
- Jaccard, P. 1912. “The Distribution of the Flora in the Alpine Zone.” *New Phytologist* 11, no. 2: 37–50. <https://doi.org/10.1111/j.1469-8137.1912.tb05611.x>.
- Jensen, M. R., E. E. Sigsgaard, M. d. P. Ávila, et al. 2022. “Short-Term Temporal Variation of Coastal Marine eDNA.” *Environmental DNA* 4, no. 4: 747–762. <https://doi.org/10.1002/edn3.285>.
- Jerde, C. L., E. A. Wilson, and T. L. Dressler. 2019. “Measuring Global Fish Species Richness With eDNA Metabarcoding.” *Molecular Ecology Resources* 19, no. 1: 19–22. <https://doi.org/10.1111/1755-0998.12929>.
- Kelly, R. P., D. M. Lodge, K. N. Lee, et al. 2024. “Toward a National eDNA Strategy for the United States.” *Environmental DNA* 6, no. 1: e432. <https://doi.org/10.1002/edn3.432>.
- Kéry, M., and B. Schmidt. 2008. “Imperfect Detection and Its Consequences for Monitoring for Conservation.” *Community Ecology* 9: 207–216. <https://doi.org/10.1556/comec.9.2008.2.10>.
- Kim, D., A. T. Taylor, and T. J. Near. 2022. “Phylogenomics and species delimitation of the economically important Black Basses (Micropterus).” *Scientific Reports* 12, no. 1: 9113. <https://doi.org/10.1038/s41598-022-11743-2>.
- Kirtane, A., J. D. Atkinson, and L. Sassoubre. 2020. “Design and Validation of Passive Environmental DNA Samplers Using Granular Activated Carbon and Montmorillonite Clay.” *Environmental Science & Technology* 54, no. 19: 11961–11970. <https://doi.org/10.1021/acs.est.0c01863>.
- Klymus, K. E., C. A. Richter, D. C. Chapman, and C. Paukert. 2015. “Quantification of eDNA Shedding Rates from Invasive Bighead Carp *Hypophthalmichthys nobilis* and Silver Carp *Hypophthalmichthys molitrix*.” *Biological Conservation* 183: 77–84. <https://doi.org/10.1016/j.biocon.2014.11.020>.
- Knight, C. L. 1987. *Fishes of the Tangipahoa River system in Southwest Mississippi and Southeast Louisiana*. M.S., Southeastern Louisiana University. <https://www.proquest.com/docview/303546053/citation/77C4C8C277F24872PQ/1>.
- Krolow, A. D., A. D. Geheber, and K. R. Piller. 2022. “If You Build It, Will They Come? An Environmental DNA Assessment of Fish Assemblages on Artificial Reefs in the Northern Gulf of Mexico.” *Transactions of the American Fisheries Society* 151, no. 3: 297–321. <https://doi.org/10.1002/tafs.10352>.
- Lamb, P. D., V. G. Fonseca, D. L. Maxwell, and C. C. Nnanatu. 2022. “Systematic Review and Meta-Analysis: Water Type and Temperature Affect Environmental DNA Decay.” *Molecular Ecology Resources* 22, no. 7: 2494–2505. <https://doi.org/10.1111/1755-0998.13627>.
- Lance, R., K. Klymus, C. Richter, et al. 2017. “Experimental Observations on the Decay of Environmental DNA from Bighead and Silver Carps.” *Management of Biological Invasions* 8, no. 3: 343–359. <https://doi.org/10.3391/mbi.2017.8.3.08>.
- Laporte, M., E. Reny-Nolin, V. Chouinard, et al. 2021. “Proper Environmental DNA Metabarcoding Data Transformation Reveals Temporal Stability of Fish Communities in a Dendritic River System.” *Environmental DNA* 3, no. 5: 1007–1022. <https://doi.org/10.1002/edn3.224>.
- Leray, M., J. Y. Yang, C. P. Meyer, et al. 2013. “A New Versatile Primer Set Targeting a Short Fragment of the Mitochondrial COI Region for Metabarcoding Metazoan Diversity: Application for Characterizing Coral Reef Fish Gut Contents.” *Frontiers in Zoology* 10, no. 1: 34. <https://doi.org/10.1186/1742-9994-10-34>.
- Lodge, D. M. 2022. “Policy Action Needed to Unlock eDNA Potential.” *Frontiers in Ecology and the Environment* 20, no. 8: 448–449. <https://doi.org/10.1002/fee.2563>.
- Louisiana Department of Wildlife and Fisheries. 2022. *Tangipahoa River Watershed, Louisiana Waterbody Management Plan—Part B Evaluation & Recommendations*. Office of Fisheries Inland Fisheries Section. https://www.wlf.louisiana.gov/assets/Resources/Publications/Freshwater_Inland_Fish/Inland-Waterbody-Management-Plans/Tangipahoa-River-MP-B-2022.pdf.
- Magoč, T., and S. L. Salzberg. 2011. “FLASH: Fast Length Adjustment of Short Reads to Improve Genome Assemblies.” *Bioinformatics* 27, no. 21: 2957–2963. <https://doi.org/10.1093/bioinformatics/btr507>.
- McCartin, L. J., S. A. Vohsen, S. W. Ambrose, et al. 2022. “Temperature Controls eDNA Persistence across Physicochemical Conditions in Seawater.” *Environmental Science & Technology* 56, no. 12: 8629–8639. <https://doi.org/10.1021/acs.est.2c01672>.
- McMurdie, P. J., and S. Holmes. 2013. “phyloseq: An R Package for Reproducible Interactive Analysis and Graphics of Microbiome Census Data.” *PLoS One* 8, no. 4: e61217. <https://doi.org/10.1371/journal.pone.0061217>.
- Miya, M., Y. Sato, T. Fukunaga, et al. 2015. “MiFish, a Set of Universal PCR Primers for Metabarcoding Environmental DNA From Fishes: Detection of More Than 230 Subtropical Marine Species.” *Royal Society Open Science* 2, no. 7: 150088. <https://doi.org/10.1098/rsos.150088>.
- Mossa, J., and S. R. Marks. 2011. “Pit Avulsions and Planform Change on a Mined River Floodplain: Tangipahoa River, Louisiana.” *Physical Geography* 32, no. 6: 512–532. <https://doi.org/10.2747/0272-3646.32.6.512>.
- Muller, T. A., and A. M. Simons. 2024. “Taxonomic Revision of the Pirate Perches, Aphredoderus, (Percopsiformes: Aphredoderidae) with Descriptions of Two New Species.” *Zootaxa* 5415, no. 1: 77–105. <https://doi.org/10.11646/zootaxa.5415.1.3>.
- Murphy, B. R., and D. W. Willis. 1996. *Fisheries Techniques*. 2nd ed. American Fisheries Society.
- Nielsen, L. A., and D. L. Johnson. 1983. *Fisheries Techniques*. 1st ed. American Fisheries Society.
- Oathout, C. E. 2024. “Short Term Environmental DNA Variation Using Active and Passive Sampling Techniques in Two Aquatic Ecosystems.” Master’s thesis, Southeastern Louisiana University.
- Oksanen, J., F. G. Blanchet, M. Friendly, et al. 2020. “Vegan community ecology package version 2.5-7 November 2020.”
- Pierce, C. L., A. M. Corcoran, A. N. Gronbach, S. Hsia, B. J. Mullarkey, and A. J. Schwartzhoff. 2001. “Influence of Diel Period on Electrofishing and Beach Seining Assessments of Littoral Fish Assemblages.” *North American Journal of Fisheries Management* 21, no. 4: 918–926. [https://doi.org/10.1577/1548-8675\(2001\)021%253C0918:IODPOE%253E2.0.CO;2](https://doi.org/10.1577/1548-8675(2001)021%253C0918:IODPOE%253E2.0.CO;2).
- Piggott, M. P., S. C. Banks, B. T. Broadhurst, C. J. Fulton, and M. Lintermans. 2021. “Comparison of Traditional and Environmental DNA Survey Methods for Detecting Rare and Abundant Freshwater Fish.” *Aquatic Conservation: Marine and Freshwater Ecosystems* 31, no. 1: 173–184. <https://doi.org/10.1002/aqc.3474>.
- Pompeu, P. S., D. R. de Carvalho, C. G. Leal, et al. 2021. “Sampling Efforts for Determining Fish Species Richness in Megadiverse Tropical Regions.” *Environmental Biology of Fishes* 104, no. 11: 1487–1499. <https://doi.org/10.1007/s10641-021-01184-7>.
- Port, J. A., J. L. O’Donnell, O. C. Romero-Maraccini, et al. 2016. “Assessing Vertebrate Biodiversity in a Kelp Forest Ecosystem Using Environmental DNA.” *Molecular Ecology* 25, no. 2: 527–541. <https://doi.org/10.1111/mec.13481>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Roberts, D. W. 2023. “labdsv: Ordination and Multivariate Analysis for Ecology (p. 2.1-0) [Dataset].” <https://doi.org/10.32614/CRAN.package.labdsv>.

- Robinson, A. T., Y. M. Paroz, M. J. Clement, et al. 2019. "Environmental DNA Sampling of Small-Bodied Minnows: Performance Relative to Location, Species, and Traditional Sampling." *North American Journal of Fisheries Management* 39, no. 5: 1073–1085. <https://doi.org/10.1002/nafm.10344>.
- Sato, Y., M. Miya, T. Fukunaga, T. Sado, and W. Iwasaki. 2018. "MitoFish and MiFish Pipeline: A Mitochondrial Genome Database of Fish With an Analysis Pipeline for Environmental DNA Metabarcoding." *Molecular Biology and Evolution* 35, no. 6: 1553–1555. <https://doi.org/10.1093/molbev/msy074>.
- Schmieder, R., Y. W. Lim, F. Rohwer, and R. Edwards. 2010. "TagCleaner: Identification and Removal of Tag Sequences From Genomic and Metagenomic Datasets." *BMC Bioinformatics* 11, no. 1: 341. <https://doi.org/10.1186/1471-2105-11-341>.
- Stevens, E. R., J. Hyde, L. S. Beesley, et al. 2024. "Fishy Business—Assessing the Efficacy of Active and Passive eDNA to Describe the Fish Assemblage of a River in Southwestern Western Australia to Support Effective Monitoring." *Environmental DNA* 6, no. 6: e70040. <https://doi.org/10.1002/edn3.70040>.
- Strickler, K. M., A. K. Fremier, and C. S. Goldberg. 2015. "Quantifying Effects of UV-B, Temperature, and pH on eDNA Degradation in Aquatic Microcosms." *Biological Conservation* 183: 85–92. <https://doi.org/10.1016/j.biocon.2014.11.038>.
- Takahara, T., T. Minamoto, H. Yamanaka, H. Doi, and Z. Kawabata. 2012. "Estimation of Fish Biomass Using Environmental DNA." *PLoS One* 7, no. 4: e35868. <https://doi.org/10.1371/journal.pone.0035868>.
- Tangipahoa River, La, Entrance Channel. 1970. "United States Army Corps of Engineers."
- Thomas, A. C., J. Howard, P. L. Nguyen, T. A. Seimon, and C. S. Goldberg. 2018. "eDNA Sampler: A Fully Integrated Environmental DNA Sampling System." *Methods in Ecology and Evolution* 9, no. 6: 1379–1385. <https://doi.org/10.1111/2041-210X.12994>.
- Thomsen, P. F., J. Kielgast, L. L. Iversen, P. R. Møller, M. Rasmussen, and E. Willerslev. 2012. "Detection of a Diverse Marine Fish Fauna Using Environmental DNA From Seawater Samples." *PLoS One* 7, no. 8: e41732. <https://doi.org/10.1371/journal.pone.0041732>.
- Villegas-Ríos, D., J. Alós, M. Palmer, et al. 2014. "Life-History and Activity Shape Catchability in a Sedentary Fish." *Marine Ecology Progress Series* 515: 239–250. <https://doi.org/10.3354/meps11018>.
- Wickham, H. 2011. "Ggplot2." *WIREs Computational Statistics* 3, no. 2: 180–185. <https://doi.org/10.1002/wics.147>.
- Wilcox, T. M., K. S. McKelvey, M. K. Young, et al. 2016. "Understanding Environmental DNA Detection Probabilities: A Case Study Using a Stream-Dwelling Char *Salvelinus fontinalis*." *Biological Conservation* 194: 209–216. <https://doi.org/10.1016/j.biocon.2015.12.023>.
- Yu, D. W., Y. Ji, B. C. Emerson, et al. 2012. "Biodiversity Soup: Metabarcoding of Arthropods for Rapid Biodiversity Assessment and Biomonitoring." *Methods in Ecology and Evolution* 3, no. 4: 613–623. <https://doi.org/10.1111/j.2041-210X.2012.00198.x>.
- Zale, A. V., D. L. Parrish, and T. M. Sutton. 2013. *Fisheries Techniques*. 3rd ed. American Fisheries Society. <https://doi.org/10.47886/9781934874295>.
- Zhang, L., W. Zhou, M. Jiao, et al. 2024. "Use of Passive Sampling in Environmental DNA Metabarcoding Technology: Monitoring of Fish Diversity in the Jiangmen Coastal Waters." *Science of the Total Environment* 908: 168298. <https://doi.org/10.1016/j.scitotenv.2023.168298>.
- Zhao, B., P. M. van Bodegom, and K. B. Trimbos. 2023. "Bacterial Abundance and pH Associate With eDNA Degradation in Water From Various Aquatic Ecosystems in a Laboratory Setting." *Frontiers in Environmental Science* 11. <https://doi.org/10.3389/fenvs.2023.1025105>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Rarefaction curves of read abundances for the detected fishes where the x-axis represents the read count of the sample and the y-axis represents the number of taxa detected in that sample. **Figure S2:** Changes in abiotic factors measured across the sampling period. The y-axis represents different abiotic factors; flow (m/s), temperature (°C), dissolved oxygen (mg/L and %), turbidity (NTU), pH, and salinity (ppt), while the x-axis represents the date and time of sample collection over the 25-h period. **Table S1:** The read counts of all species detected during each sampling period using aquatic eDNA as a sampling technique in the Tangipahoa River at Cherokee Beach, LA (30.602361, −90.437806). Active sampling periods, civil twilight, representing sampling hours 1 and 2 on October 22 and 16–25 on October 23, 2022 and dark, representing sampling hours 3–8 on October 22 and 9–15 on October 23, 2022. Passive sampling period day, is represented by the full 25 h of submersion from 1600 on October 22 to 1600 on October 23, 2022.